

LatticIA: Childbirth and Mother Characters in Big Literary Corpora

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This ongoing research in computational literary studies focuses on the automated detection of childbirth scenes. Using distant reading, we investigate the role childbirth plays in French literature and how it has evolved over time. We are developing a model capable of automatically detecting such scenes in order to address quantitative research questions, such as how frequently childbirth events occur in French literature across periods and genres. In addition, we conduct qualitative analyses of the detected scenes, with particular attention to mother characters and their portrayal. This abstract concentrates on the pipeline currently being developed to train the model for childbirth scene detection.

Keywords: computational literary studies, distant reading, childbirth, scene detection, literary character

1 Context

Inspired by previous studies that have highlighted stereotypes associated with women in literature [16; 22; 23], using distant reading techniques [15; 21], this project focuses on childbirth scenes and the figure of the mother.

Beyond the initial objective of quantifying the recurrence of childbirth scenes across genres and periods using large literary corpora, the study also aims to provide qualitative analyses. It examines the metaphors employed [20] and recurring topoi, such as death in childbirth¹, in order to explore how literature represents birth and motherhood. It further seeks to determine whether an archetype of the mother exists, which stereotypes are associated with it, and how these have evolved over time. This research complements existing literary studies that address these questions without a computational component [4; 10; 11; 18].

Recent studies have demonstrated the possibility of detecting prototypical scenes—such as those involving danger [1; 19]. Frameworks have also been developed to support large-scale qualitative analyses of fictional characters, for example to describe their social status or physical traits [3; 17]. The LatticIA project builds on these approaches.

As the project is still in its early stages, the remainder of this article focuses on the pipeline currently being developed for the automatic detection of childbirth scenes.

2 Method

We employ a processing pipeline whose stages are detailed in Figure 1.

Paper under review.

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¹ <https://web.archive.org/web/20180920050342/http://satorbase.org/index.php?do=topoi&topoi=10825>

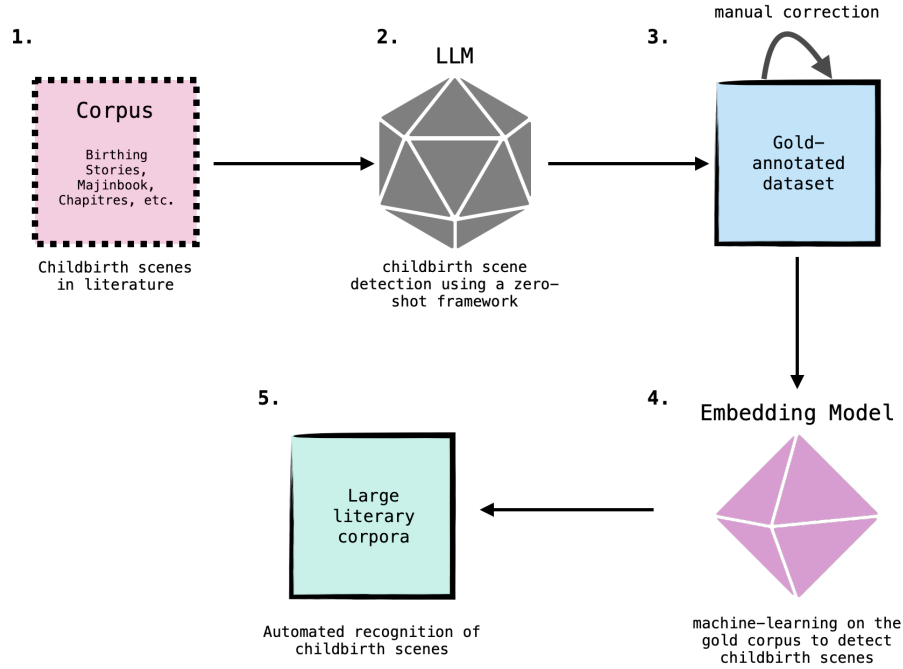


Figure 1: Overview of the Processing Pipeline.

1. First, we compile a training corpus based on the Birth(ing) Stories database [7], developed within the Birth(ing) Stories project: a project that aims of uncovering the issues of aesthetic and cultural representation surrounding childbirth in literature. It contains the titles [6] of 143 books from the 12th to the 21st century, each featuring at least one childbirth scene. This database serves as a starting point for building the training corpus. We search for full-text versions of these works in large literary corpora (Chapitres [13], Gallica, and Majinbook [14]).
2. The training corpus for the childbirth scene detection model is built using Retrieval-Augmented Generation (RAG) [9] in a zero-shot framework, following a strict set of annotation guidelines [8] developed within our project Latticia. These guidelines address the following questions:
 - Is this a childbirth scene?
 - What is the phase of labor?
 - Which characters are present?
 - Is pregnancy mentioned?
 - Is there medical intervention?
 - Are there complications?
3. We then conduct an iterative process combining manual corrections and new automated annotations using the INCEpTION semantic annotation platform [12]. At this stage, human annotators play a crucial role: the interaction between the LLM, the annotation guidelines, and inter-annotator discussions enables the refinement of a gold-standard annotated dataset.
4. Based on this gold-standard dataset, we train a classification model to automatically detect childbirth scenes across a larger corpus, using contextualized embeddings using models with a BERT-based architecture [2].

5. Finally, the classification model working with contextualized embeddings is applied to automatically identify childbirth scenes in large-scale literary corpora (Chapitres [13], Gallica, and Majinbook [14]).

3 Conclusion

Computational methods make it possible to quantify and analyze the role of childbirth—and more specifically the mother character—in literature. Within close reading, it is often evident that female characters are frequently marginalized or stereotyped, particularly in relation to childbirth [4; 11; 18]. Approaching this topic through computational methods and distant reading [15] not only offers a new perspective but also reveals long-term evolutionary patterns in literary representation and provides new analytical tools. A seemingly minor element in a single work can acquire systematic significance when examined at scale.

The stereotypes and representations found in literature reveal much about how childbirth and the role of the mother have been perceived in society. Beliefs about childbirth profoundly influence women’s personal experiences [5]. By identifying large-scale patterns in literary representations, this study aims to contribute to a clearer understanding of the origins and persistence of ideas surrounding childbirth.

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